



COURSE DESCRIPTION FORM: CS-2009: Design and Analysis of Algorithms

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad Campus

PROGRAM TO BE EVALUATED BS-CS: Fall-2025

Course Description

Course Code	CS-2009	
Course Title	Design and Analysis of Algorithms	
Credit Hours	3	
Course Instructors	Noor ul Ain, Owais Idrees, Hira Mastoor	
Grading Policy	Absolute grading	
Policy about missed assessment items in the course	Retake of missed assessment items (other than sessional/ final exam) will not be held. Student who misses an assessment item (other than sessional / final exam) is awarded zero marks in that assessment item i.e. late submission will not be accepted. For missed sessional/ final exam, exam retake/ pretake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decides the exam retake/ pretake cases.	
Course Plagiarism Policy	Plagiarism in project or sessional/ final exam will result in F grade in the course. Plagiarism in an assignment/ quiz will result in zero marks in the whole assignments/ quizzes category.	
Prerequisites by Course(s) or Topics	Data Structures	
Assessment Instruments with Weights (homeworks, quizzes, sessional exams, final exam, assignments, etc.)	Assessment with the weight.	
	Assessment Type	Weight
	Assignments (3)	10
	Quizzes	10
	Project	10
	Sessional Exams (2)	30
	Final Exam	40
Course Coordinator	Noor ul Ain	
URL (if any)		
Course Catalog Description	Introduction; role of algorithms in computing, Analysis on nature of input and size of input Asymptotic notations; Big-O, Big Ω , Θ , little-o, little- ω , Sorting Algorithm analysis, loop invariants, Recursion and recurrence relations; Algorithm Design Techniques, Brute Force Approach, Divide-and-conquer approach; Merge, Quick Sort, Greedy approach; Dynamic	

	programming; Elements of Dynamic Programming, Search trees; Heaps; Hashing; Graph algorithms, shortest paths, sparse graphs, String matching, Amortized Analysis																
Textbook	Thomas H. Cormen et al. "Introduction to Algorithms" 3 rd Edition.																
Reference Material	Anany Levitin "Introduction to the design and analysis of algorithms" 3 rd Edition.																
Course Goals	A. Course Learning Outcomes (CLOs) After course completion, the students shall be able to: 1. Design algorithms using different algorithms design techniques i.e. Brute Force, Divide and Conquer, Dynamic Programming, Greedy Algorithms and apply them to solve problems in the domain of the program (6) (4) 2. Analyze the time and space complexity of different algorithms by using standard asymptotic notations for recursive and nonrecursive algorithms.(4) (3) 3. Describe, compare, analyze, and solve general algorithmic problem types: Sorting, Searching, String Processing, Graph. 4. Evaluate the correctness of algorithms by using theorem proving or executing test cases (5) (3) 5. Implement the algorithms, compare the implementations empirically, and apply fundamental algorithms knowledge to solve practical problems related to the program. (3) (4) B. Program Learning Outcomes (PLOs) <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 5%; text-align: center;">1</td><td style="width: 45%;">Academic Education</td><td>Completion of an accredited program of study designed to prepare graduates as computing professionals</td></tr> <tr> <td style="text-align: center;">2</td><td>Knowledge for Solving Computing Problems</td><td>Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements</td></tr> <tr> <td style="text-align: center;">3</td><td>Problem Analysis</td><td>Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines</td></tr> <tr> <td style="text-align: center;">4</td><td>Design/ Development of Solutions</td><td>Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations</td></tr> <tr> <td style="text-align: center;">5</td><td>Modern Tool Usage</td><td>Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations</td></tr> </table>		1	Academic Education	Completion of an accredited program of study designed to prepare graduates as computing professionals	2	Knowledge for Solving Computing Problems	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements	3	Problem Analysis	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines	4	Design/ Development of Solutions	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations	5	Modern Tool Usage	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations
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	6	Individual and Team Work	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings									
	7	Communication	Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions									
	8	Computing Professionalism and Society	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice									
	9	Ethics	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice									
	10	Life-long Learning	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional									
	C. Mapping of CLOs to PLOs (CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)											
		PLOs										
			1	2	3	4	5	6	7	8	9	10
CLOs	1	☐	☐		☐		☐					
	2	☐	☐	☐			☐					
	3	☐	☐	☐			☐					
	4	☐	☐		☐		☐					
	5	☐	☐		☐		☐					
	6	☐	☐	☐			☐					
Topics covered in the course (assume 15-week instruction and 3 contact hours per week)	1. Topics to be covered:											
	List of Topics					No. of Weeks	Contact Hours	CLO(s)				
	Basics of Algorithms, Mathematical Foundation, Growth of Function, Asymptotic Notations.					2	6	2,3				
	Divide and Conquer, Recurrences,					2	6	2,3				

	Iteration/ algebraic method, Recurrence-Tree Method, Master's Method.					
	Sorting (Merge, Insertion, Quick, Heap, Counting, Radix), Heaps		2	6	1,2,3,4	
	String Matching Algorithms		2	6	2,4,5	
	Greedy Algorithms (Graph Terminology, Representation of Graphs, BFS & DFS, Kruskal's Algorithm, Prim's Algorithm, Dijkstra's Algorithm)		1.5	4.5	2, 4	
	Dynamic Programming		4.5	13.5	1,2	
	Hashing		0.5	1.5	1,4,6	
	NP Complete Problems and Solutions		0.5	1.5	1,4,6	
	Total		15	45		
	Programming Language for Assignments	C++				
Class Time Spent (in percentage)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues		
	55	20	20	5		
Oral and Written Communications	Every student is required to submit at least __5__ written reports of typically __6__ pages each.					